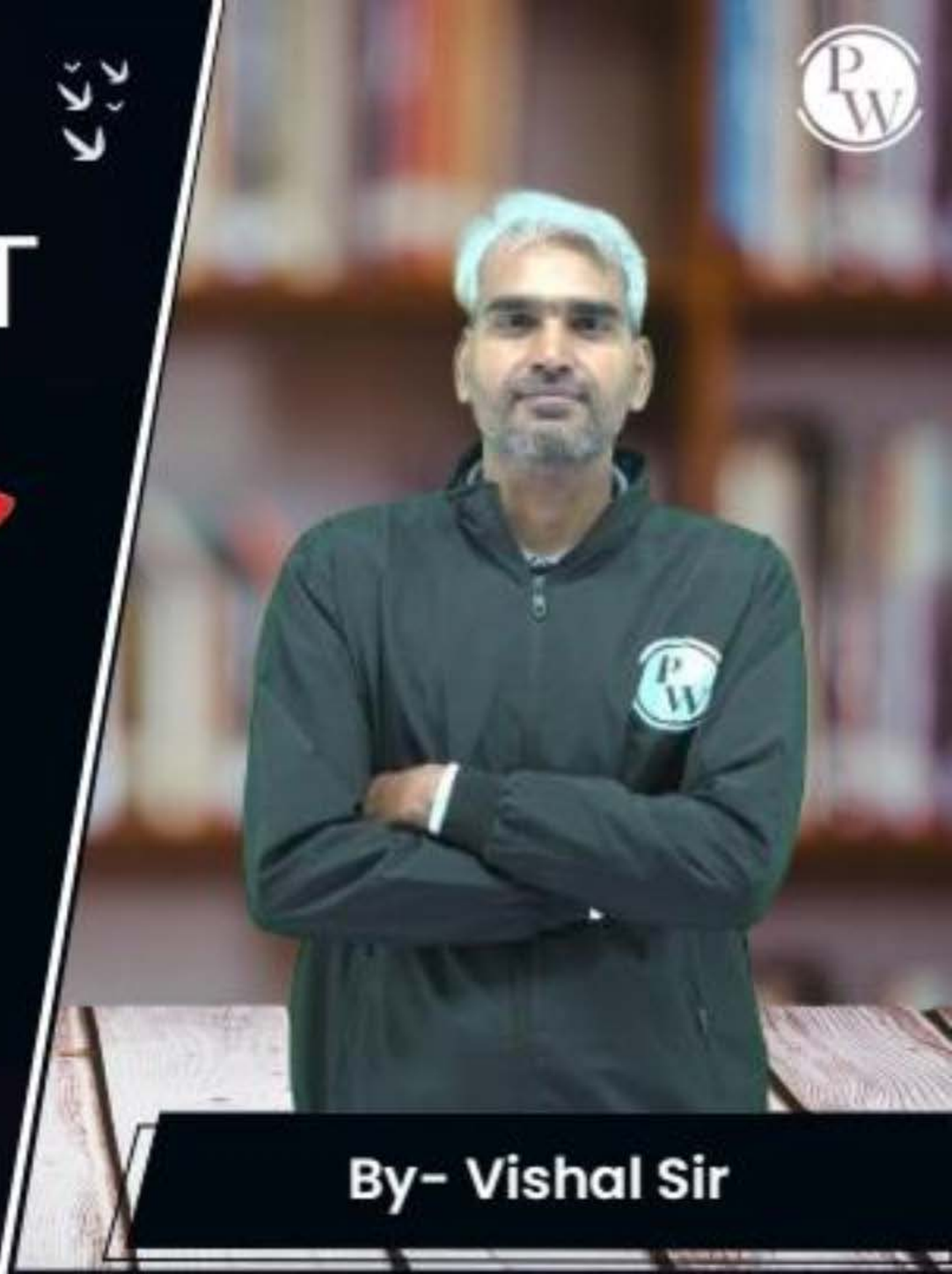
Computer Science & IT



Compiler Design

Syntax Analysis (Parser)

Lecture No. 13



By- Vishal Sir

Recap of Previous Lecture



Topic Need of LR(1) items

Topic Construction of DFA of LR(1) items

Slide

Topics to be Covered

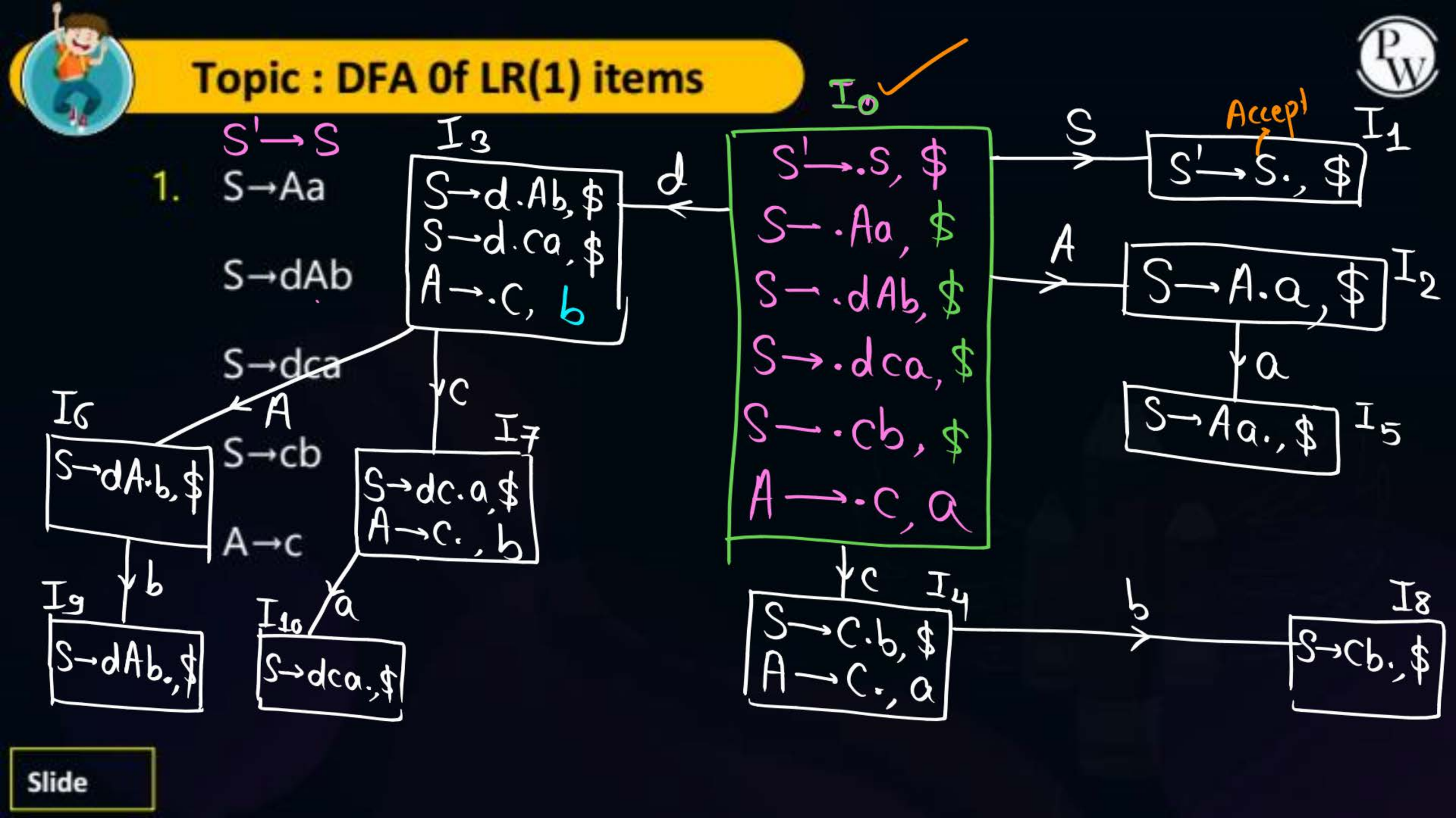


Topic — Construction of DFA of LR(1) items

Topic — CLR(1) Parser

Slide

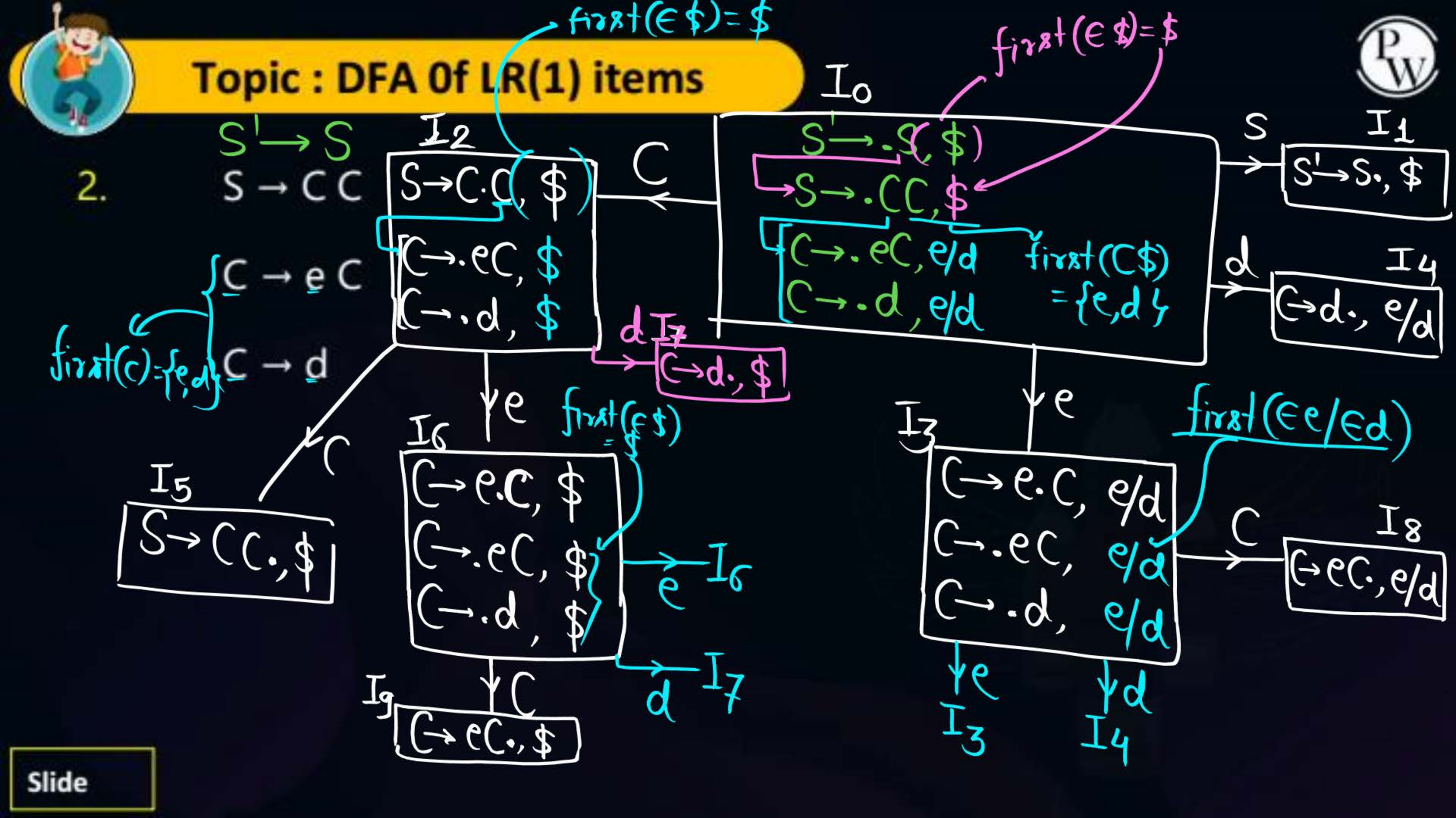
Topic : DFA Of LR(1) items



Slide

Topic : DFA Of LR(1) items

2. $S' \to S$
$S \to CC$
$C \to eC$
$C \to d$



Slide

Topic : DFA Of LR(1) items

3.

$S' \rightarrow S$

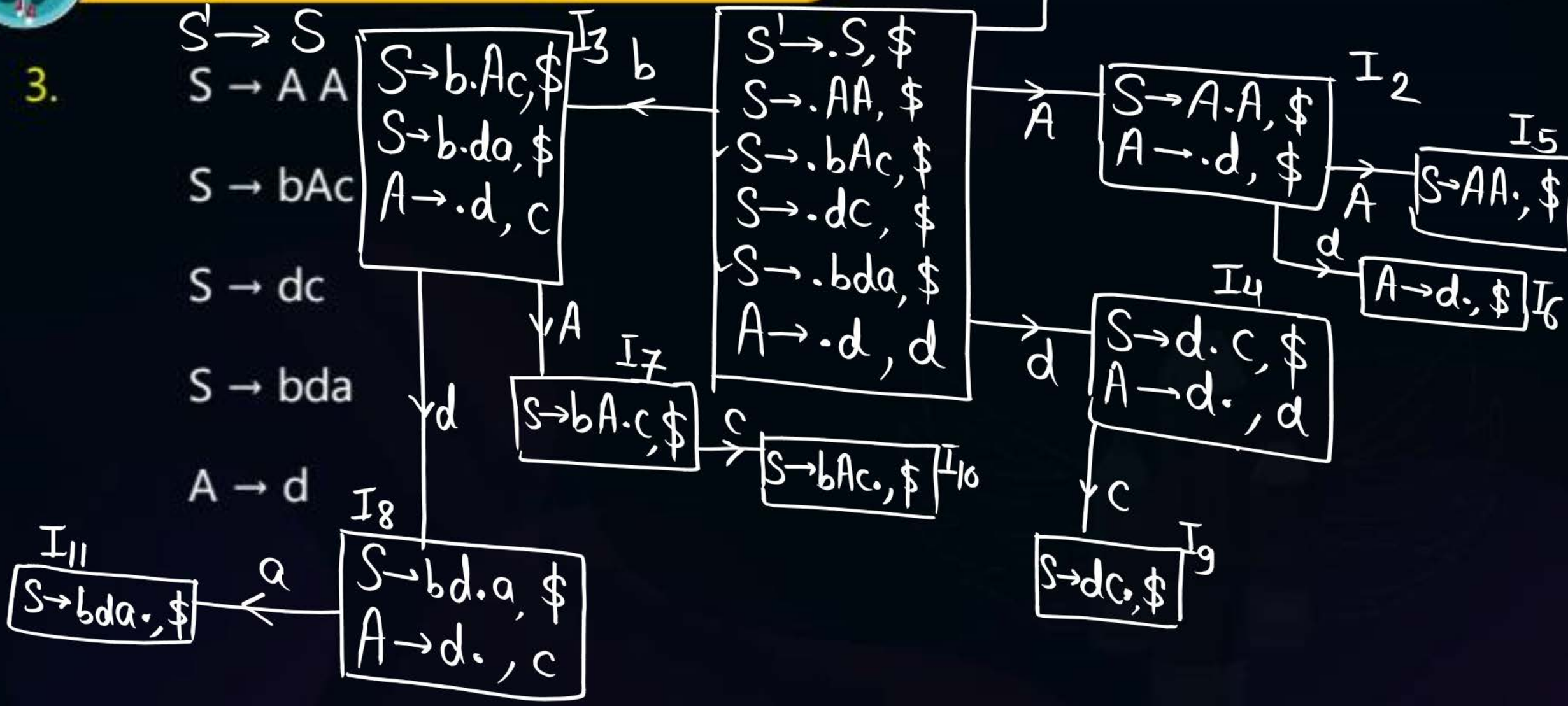
$S \rightarrow AA$

$S \rightarrow bAc$

$S \rightarrow dc$

$S \rightarrow bda$

$A \rightarrow d$

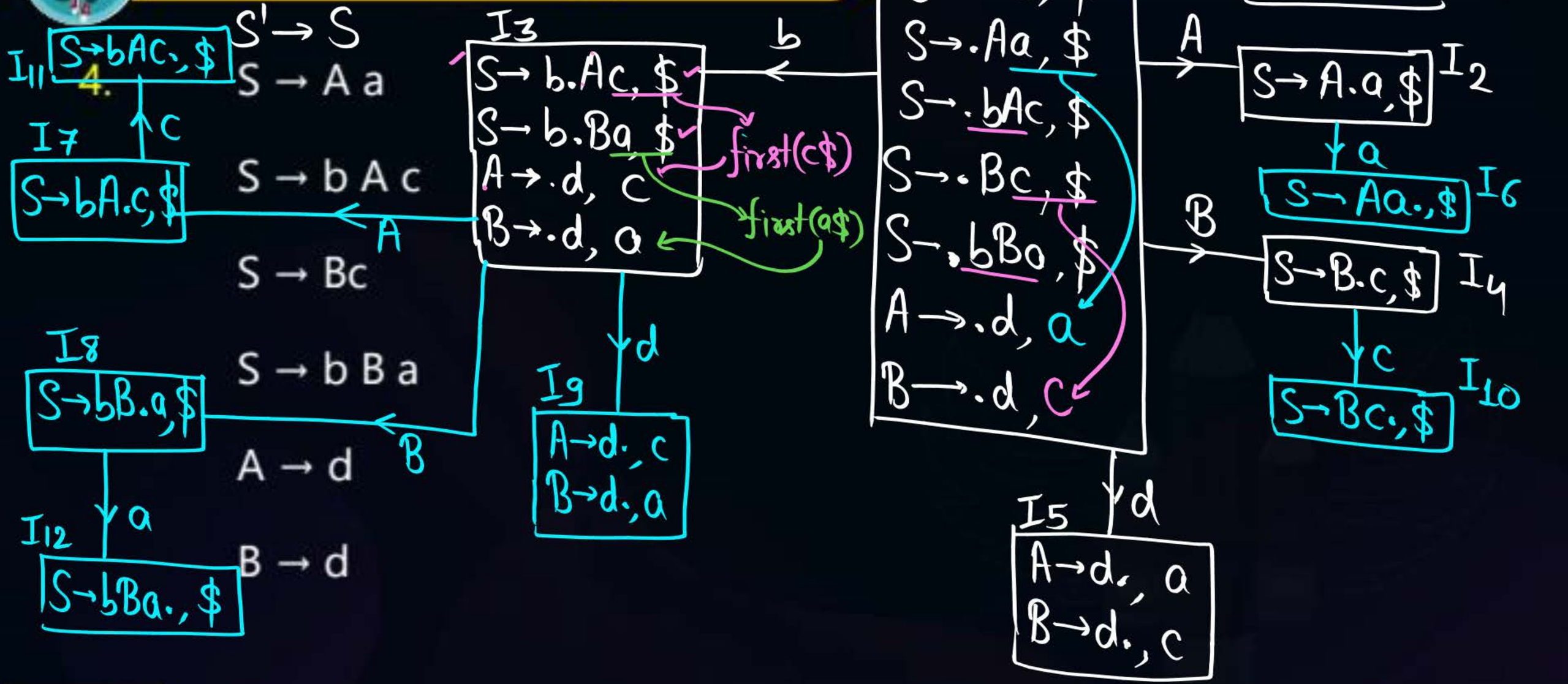


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Topic : DFA Of LR(1) items

S' → S
S → A a
S → b A c
S → Bc
S → b B a
A → d
B → d



Slide

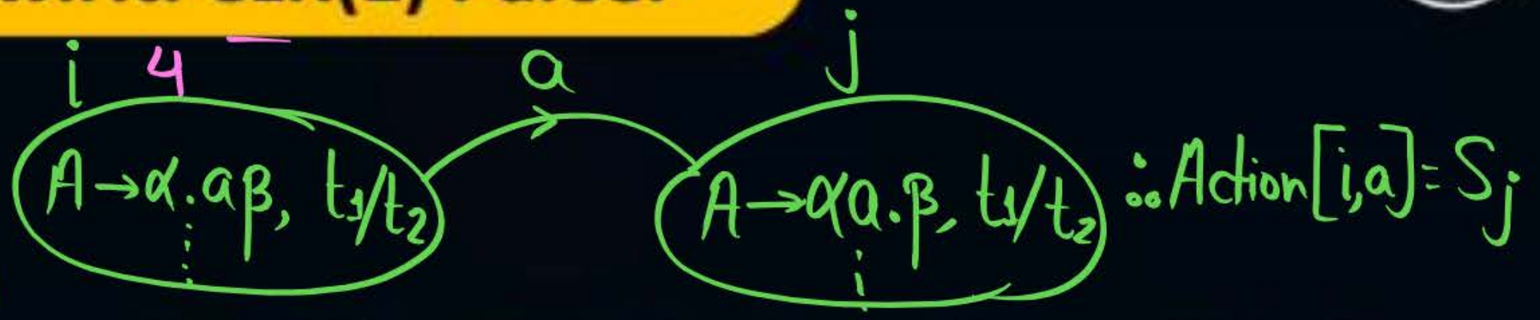


Topic : CLR(1) Parser

1. Obtain the augmented grammar.
2. Construct the DFA of LR(1) items.
3. Construct CLR(1) Parser table using the following rules

Topic : ACTION Part w.r.t. CLR(1) Parser

1. Shift Action : Same as LR(0)



∴ Action[i,a] = Sj

2. Reduce Action : If state "i" of the dfa includes an item of the form "A→α . , t1/t2...", where A→α is the jth production of grammar, then update

ACTION [i , n] = Rj For all n ∈{t1,t2,...}

Reduce entry w.r.t. all the terminals in look-ahead part.

3. Accept : i S'→S., $ ∴ Action[i,$] = Accept

4. Error : All other entries are error

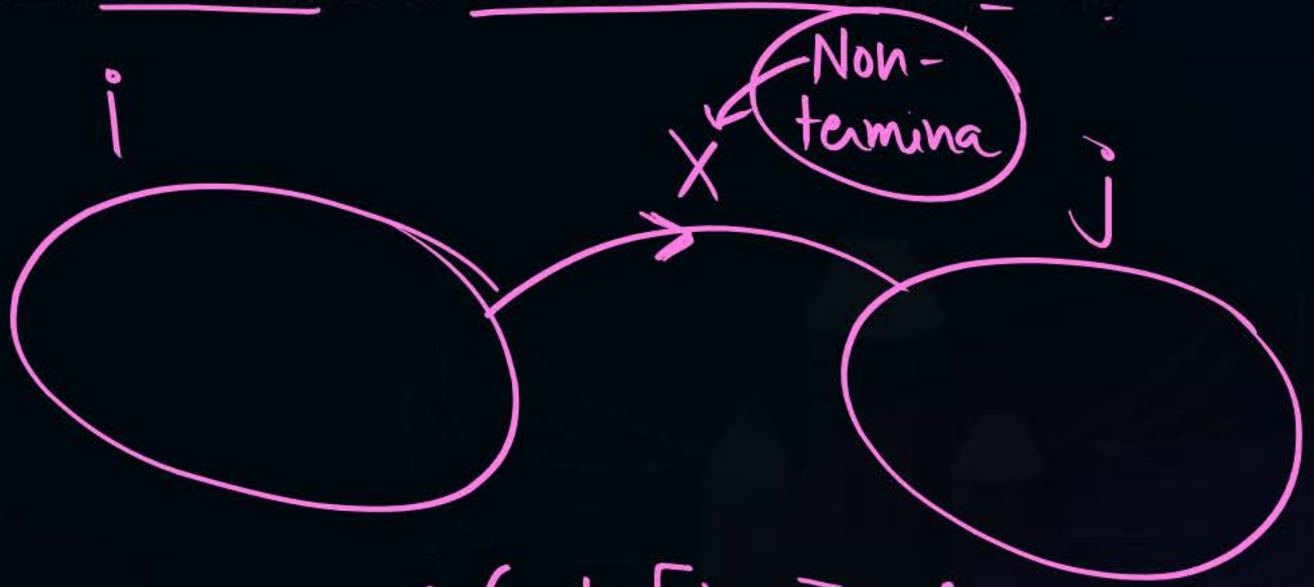
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Topic : GOTO Part w.r.t. CLR(1) Parser

- If there exists a transition out of state "i" on non-termina X to state "j" , then

  GOTO [i , X] = j



∴ Goto [i, X] = j

Slide

Topic : Construction of CLR(1) Parser table

Construct the CLR(1) parsing table for the grammar.

① S→Aa

② S→dAb

③ S→dca

④ S→cb

⑤ A→c

	ACTION					GOTO	
State	a	b	c	d	$	S	A
0			S_4	S_3		1	2
1					Accept		
2	S_5						
3			S_7				6
4	R_5	S_8					
5					R_1		
6		S_9					
7	S_{10}	R_5					
8					R_4		
9					R_2		
10					R_3		

Slide



Topic : CLR(1) Grammar

If no entry in Action part of CLR(1) Parser table contains more than one option, then Corresponding Grammar is CLR(1) Grammar

* If Any entry in Action part of CLR(1) parser table Contains multiple option then Grammar is not CLR(1) Grammar

Slide



Topic : CLR(1) Grammar

(i.e. State with SR or RR Conflict)

If no inadequate state exists in the DFA of LR(1) items with respect to CLR(1) Parser, then grammar is CLR(1) grammar.

(State with SR or RR Conflict)

If any inadequate state exists in the DFA of LR(1) items with respect to CLR(1) Parser, then grammar is not CLR(1) grammar.



Topic : SR Conflict w.r.t CLR(1) Parser

- SR Conflict exists in state 'i' w.r.t. CLR(1) Parser only if, state 'i' Contains LR(1) items of the form 'i'

terminal.

$A \longrightarrow \alpha . a\beta$, $t_1/t_2/\ldots$

$B \longrightarrow \gamma .$, $t_1'/t_2'/\ldots$

and

$a \in \{t_1', t_2', \ldots\}$

Slide



Topic : RR Conflict w.r.t CLR(1) Parser

RR Conflict exists in state 'i' w.r.t. CLR(1) Parser only if state 'i' contains LR(1) items of the form

i

$A \longrightarrow \alpha.$, $t_1/t_2 \ldots$

$B \longrightarrow \beta.$, $t_1'/t_2' \ldots$

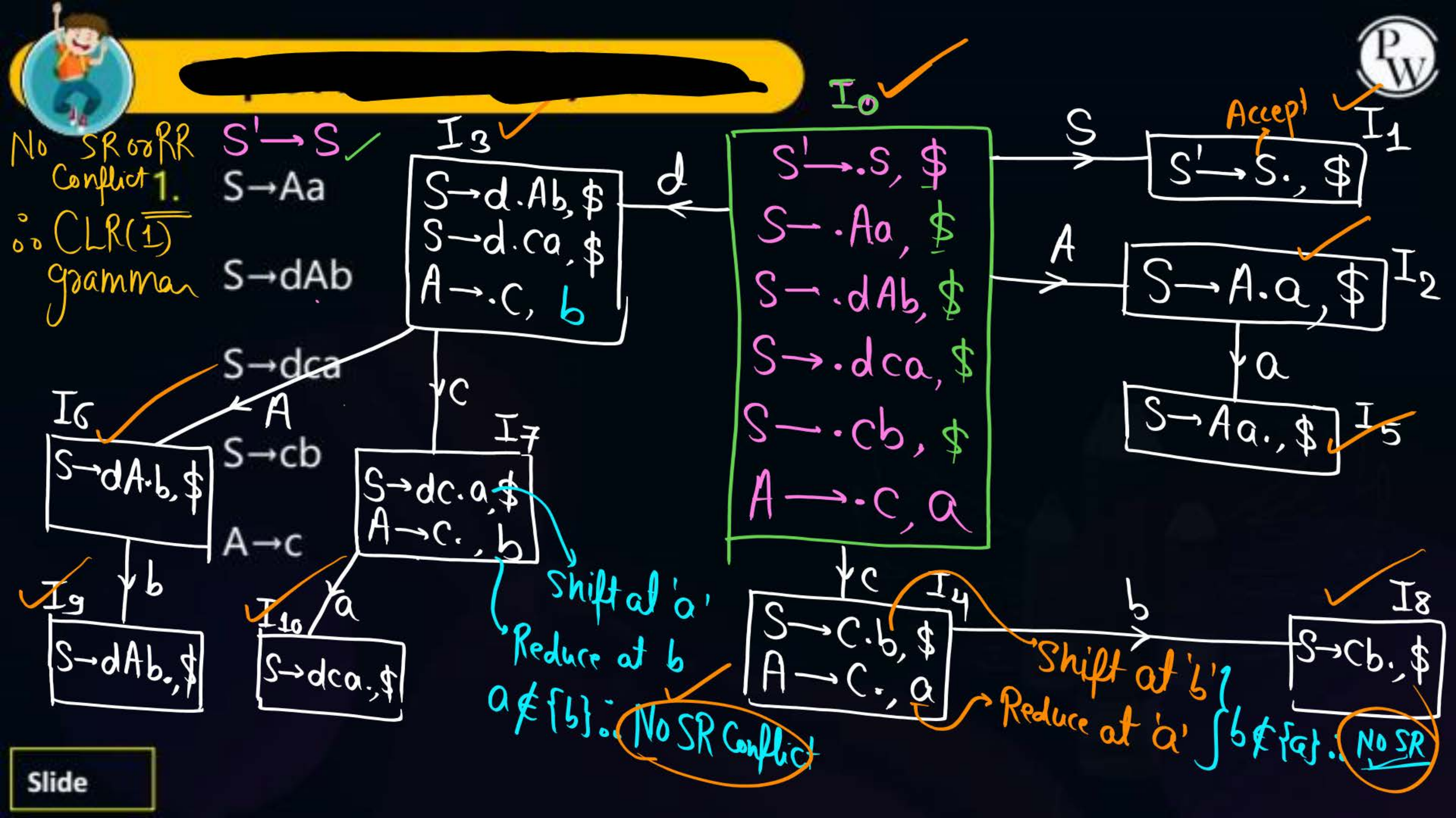
Reduce entry at terminals $t_1, t_2, \ldots$

Reduce entry at terminals $t_1', t_2', \ldots$

and,

$\{t_1, t_2, \ldots\} \cap \{t_1', t_2', \ldots\} \neq \emptyset$

Slide


No SR or RR Conflict
∴ CLR(1) grammar
S'→S
1. S→Aa
S→dAb
S→dca
S→cb
A→c
I0
S'→.S, $
S→.Aa, $
S→.dAb, $
S→.dca, $
S→.cb, $
A→.c, a
I1
S'→S., $
Accept
I2
S→A.a, $
I3
S→d.Ab, $
S→d.ca, $
A→.c, b
I4
S→c.b, $
A→c., a
I5
S→Aa., $
I6
S→dA.b, $
I7
S→dc.a, $
A→c., b
I8
S→cb., $
I9
S→dAb., $
I10
S→dca., $
Shift at 'a'
Reduce at b
a ∉ {b} ∴ No SR Conflict
Shift at 'b'
Reduce at 'a'
b ∉ {a} ∴ No SR
Slide

Topic : DFA Of LR(1) items



2.

$S' \to S$

① $S \to CC$

② $C \to eC$

③ $C \to d$

first(C) = {e, d}

first(∈$) = $

first(∈$) = $

first(C$) = {e, d}

first(∈e/∈d)

I_0

$S' \to .S, \$$

$S \to .CC, \$$

$C \to .eC, e/d$

$C \to .d, e/d$

I_1

$S \to S', \$$

I_2

$S \to C.C, \$$

$C \to .eC, \$$

$C \to .d, \$$

I_3

$C \to e.C, e/d$

$C \to .eC, e/d$

$C \to .d, e/d$

I_4

$C \to d., e/d$

I_5

$S \to CC., \$$

I_6

$C \to e.C, \$$

$C \to .eC, \$$

$C \to .d, \$$

I_7

$C \to d., \$$

I_8

$C \to eC., e/d$

I_9

$C \to eC., \$$

	e	d	$
4	R_3	R_3	

No SR or RR Conflict w.r.t CLR(1) .. CLR(1) grammar

Slide

Topic : DFA Of LR(1) items

3.

$S' \to S$

$S \to AA$

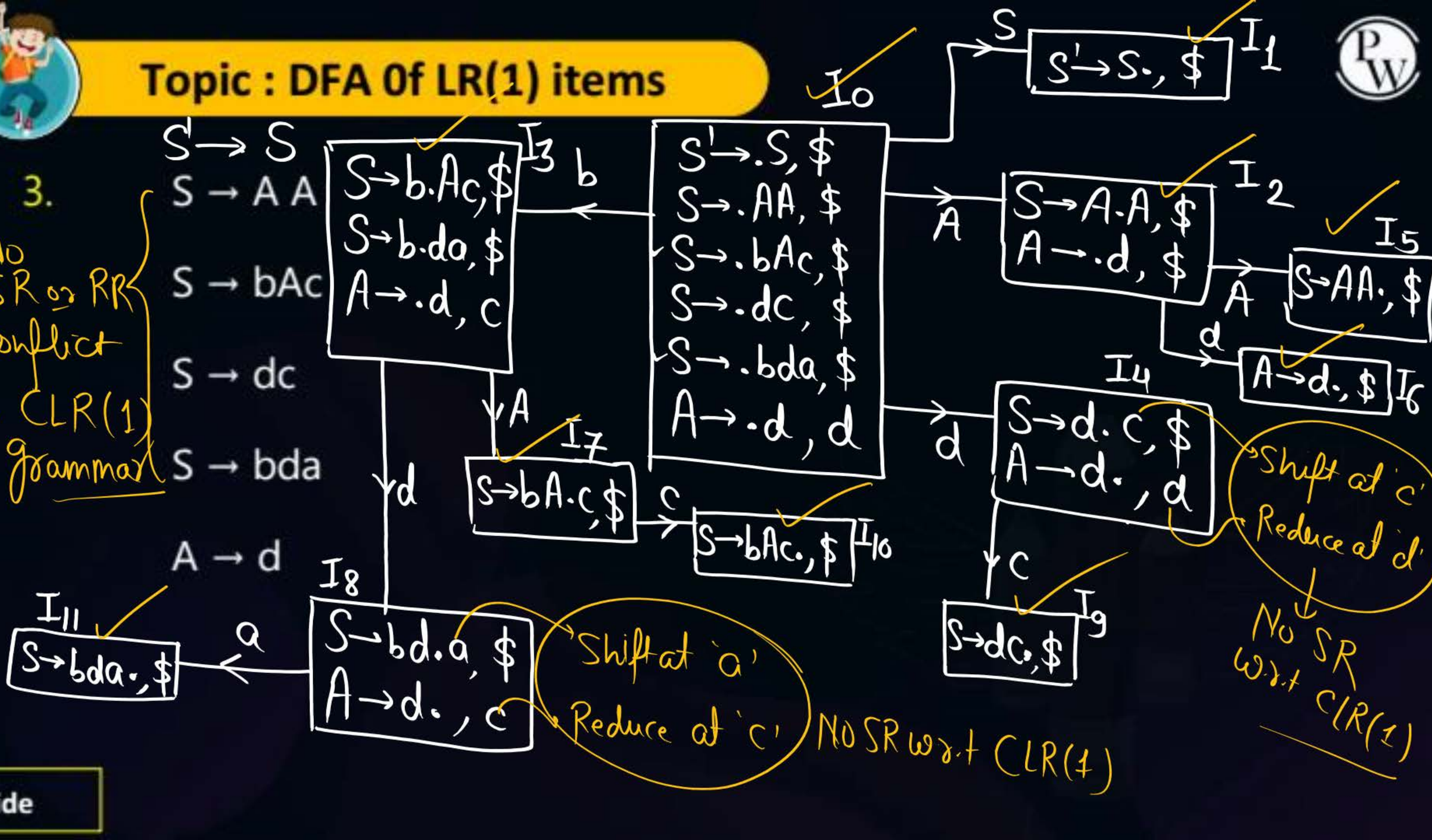
$S \to bAc$

$S \to dc$

$S \to bda$

$A \to d$

NO SR or RR Conflict ∴ CLR(1) grammar



Slide

Topic : DFA Of LR(1) items



I_0
$S' \to .S, \$$
$S \to .Aa, \$$
$S \to .bAc, \$$
$S \to .Bc, \$$
$S \to .bBa, \$$
$A \to .d, a$
$B \to .d, c$

S → I_1: $S' \to S., \$$

A → I_2: $S \to A.a, \$$

a → I_6: $S \to Aa., \$$

B → I_4: $S \to B.c, \$$

c → I_{10}: $S \to Bc., \$$

b → I_3:
$S \to b.Ac, \$$
$S \to b.Ba, \$$
$A \to .d, c$
$B \to .d, a$

first(c$)

first(a$)

d → I_9:
$A \to d., c$
$B \to d., a$

Reduce at 'c'

Reduce at 'a'

NO RR Conflict

d → I_5:
$A \to d., a$
$B \to d., c$

Reduce at 'a'

Reduce at 'c'

{a} ∩ {c} = ∅ ∴ No RR Conflict

I_{11}: $S \to bAc., \$$ (4.)

c → I_7: $S \to bA.c, \$$

A → I_{11}: $S \to bAc., \$$

a → I_{12}: $S \to bBa., \$$

B → I_8: $S \to bB.a, \$$

$S' \to S$
$S \to Aa$
$S \to bAc$
$S \to Bc$
$S \to bBa$
$A \to d$
$B \to d$

NO SR or RR Conflict ∴ CLR(1)

Slide



Topic : Parsing algorithm w.r.t CLR(1)

* LR parsing algorithm is same for all LR parsers.

Slide

Note: ① By default LR(1) parser is CLR(1) parser.

* ② Core-set of LR(1) item

In LR(1) the LR(0) part is called Core-set of LR(1) item

LR(1) item = LR(0) item + look-ahead of length '1'

eg of LR(1) item = $[A \rightarrow \alpha . \beta, t_1]$

Core-set



Topic : Note w.r.t. LR(1) items

* For CLR(1) parser we discard LR(0) item set, and we use LR(k) items {where k = 1, 2, 3, ...}, as we increase the value of 'k' the number of states in the DFA of LR(k) items may increase exponentially, ∴ we use k = 1, and item becomes LR(1) item.

Slide



2 mins Summary

Topic Construction of DFA of LR(1) items

Topic CLR(1) Parser

Topic CLR(1) Grammar



THANK - YOU

Slide